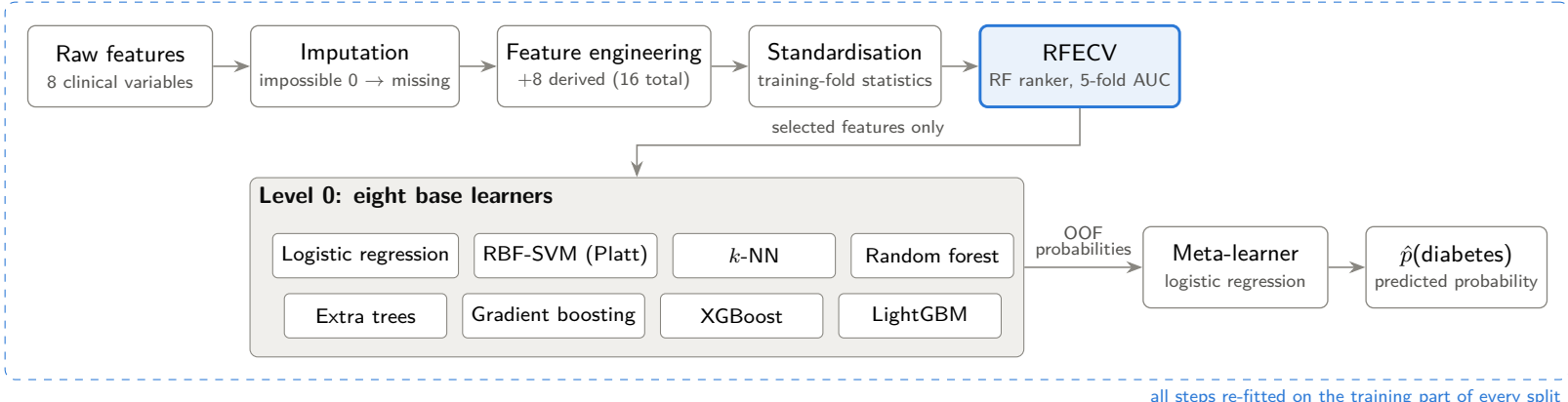

Raw features
8 clinical variables
Imputation
impossible 0 → missing
Feature engineering
+8 derived (16 total)
Standardisation
training-fold statistics
RFECV
RF ranker, 5-fold AUC
selected features only
Level 0: eight base learners
Logistic regression
RBF-SVM (Platt)
k-NN
Random forest
Extra trees
Gradient boosting
XGBoost
LightGBM
OOF
probabilities
Meta-learner
logistic regression
p̂(diabetes)
predicted probability
all steps re-fitted on the training part of every split